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Meta-Prompting and Reasoning-Based LLM Strategies for Sustainable Vehicle Telematics Optimization

by

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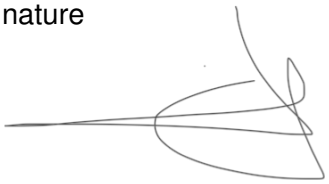
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Abstract

Vehicle telematics systems generate continuous streams of operational data covering energy consumption, driving, battery status, and vehicle dynamics. As large language models are increasingly applied to such data, two practical questions arise: which prompting strategy produces the most accurate causal discovery results, and does the choice of strategy also influence the sensitive information the model reveals in its output?

This study compares five prompting strategies: Few-Shot, Chain-of-Thought, Style, ReAct, and Meta, across three open-weight language models: Llama 3.1 (8B), DeepSeek-R1 (8B), and Qwen3 (14B). The analytical task is causal graph discovery on EV telematics dataset from the Caruso platform, evaluated against a literature-grounded ten-edge reference graph using eight structural metrics. The Sensitivity Leakage Rate (SLR) was introduced as a new binary indicator for prompt-level sensitivity leakage, enabling consistent comparisons across all conditions.

Results show that Llama 3.1 8B is the only model that has consistent non-zero causal outputs across all five strategies. Chain-of-Thought achieves the highest F1 score of 0.375, and Meta Prompting achieves a perfect precision of 1.000. DeepSeek-R1 produces empty outputs under all five thesis strategies due to an architectural mismatch: its reasoning is processed in internal thinking tokens that are removed before the output is saved. When evaluated under Step-Back Prompting outside the thesis scope, DeepSeek-R1 achieves F1 = 0.750, confirming that its causal capability exists but requires explicit output scaffolding. Qwen3 (14 billion parameters) fails to produce structured graph notation under any strategy, demonstrating that the parameter count does not determine the structured output capability.

Sensitivity leakage was not detected in any of the conditions (SLR = 0 across all conditions). A validation experiment confirmed the detection framework is sensitive: a modified open-ended prompt produced SLR = 1 uniformly across all strategies on Llama 3.1. The findings indicate that the scope of the prompt not the strategy type is predominantly responsible for driving leakage.

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List of Abbreviations

AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
RLHF	Reinforcement Learning from Human Feedback
SFT	Supervised Fine-Tuning
CoT	Chain-of-Thought
ReAct	Reasoning and Acting
SoC	State of Charge
EV	Electric Vehicle
HEV	Hybrid Electric Vehicle
BEV	Battery Electric Vehicle
NEV	New Energy Vehicle
OBD	On-Board Diagnostics
PdM	Predictive Maintenance
PHM	Prognostics and Health Management
KG	Knowledge Graph
RAG	Retrieval-Augmented Generation
LSTM	Long Short-Term Memory
RMSE	Root Mean Square Error
SLR	Sensitivity Leakage Rate
SHD	Structural Hamming Distance
FDR	False Discovery Rate
TP	True Positive
FP	False Positive
FN	False Negative
DP	Differential Privacy
GDPR	General Data Protection Regulation
RQ	Research Question
BV	Battery Voltage
SOCH	State of Charge High Level
ECA	Energy Consumption Average
ER	Electric Remaining
MIL	Mileage
SPD	Speed
ACC	Acceleration

1 Introduction

1.1 Background

Large Language Models (LLMs) have significantly changed the fields of artificial intelligence (AI) and natural language processing (NLP) by leveraging vast training corpora and deep neural architectures. Models such as GPT-4, Gemini, and LLaMA demonstrate remarkable capabilities across a wide range of tasks, from text generation and logical reasoning to summarization, translation, and domain-specific tasks. Their influence extends beyond conversational interfaces to emerging applications in scientific research, healthcare, legal analysis, engineering, and data-driven decision-making.

These achievements can be attributed to Vaswani et al. [27], who introduced the Transformer architecture. Transformers rely on self-attention mechanisms that allow the entire input sequence to be processed in parallel. Therefore, this design improves both contextual understanding and computational scalability. As model sizes grew to hundreds of billions of parameters, the term “Large Language Model” came into use to distinguish these systems from their modest-sized predecessors.

Modern Large Language Models (LLMs) are typically developed in two phases. In the pre-training phase, a model is exposed to large-scale text data in a self-supervised manner, which acquires broad linguistic competence across syntax, semantics, and context reasoning. This phase builds general capability but does not ensure alignment with specific tasks or human expectations regarding safety and usefulness. The fine-tuning phase involves subsequent strategies, including supervised fine-tuning (SFT), instruction tuning, and reinforcement learning from human feedback (RLHF). RLHF uses human preference signals to steer model outputs away from harmful, biased, or misleading content [15].

In-context learning through prompting is arguably the most important skill to acquire. By providing precisely structured natural language instructions, occasionally with a few demonstration examples, users can guide LLMs to perform new tasks [4]. However, this also introduces risks that are not well understood yet. If sensitive information is included in a prompt, the model may process or infer it in ways that may compromise user privacy. Moreover, as mentioned above, prompting is a flexible strategy, and there are some problems that need to be solved in practice, especially in analytical fields.

1.2 LLMs for Causal Discovery in Vehicle Telematics

There is growing interest in the application of LLMs to engineering and operational data domains. One particularly promising and challenging area is vehicle telematics. The current generation of automobiles creates huge amounts of data points at very high frequencies, which can be in the form of engine diagnostics, energy consumption, drivers, geographical locations, and status reports. The amount of information available makes it impossible for any human to analyze manually, necessitating the use of LLMs for this purpose.

One of the most promising applications of LLMs in telematics is causal discovery by identifying variables that drive changes in others. The causal structure of telematics variables and how they influence each other directly is relevant for predictive maintenance and energy optimization decisions. Traditional causal discovery algorithms operate on numerical data. LLMs provide an alternative; they take structured data as input and produce outputs of causal relationships. This presents a basic point not addressed in

the literature: which prompting strategy produces the most accurate and reliable causal outputs?

1.3 Problem Statement

Despite the growing use of LLMs for structured data analysis, there is no systematic understanding of how prompting affects the quality of causal discovery outputs. Studies using LLMs for vehicle maintenance and telematics have been concerned with demonstrating the model's ability to generate useful analytical outputs, but have not considered which strategy is most effective, how the choice of strategy interacts with model architecture, or whether results are consistent across repeated evaluations. This gap is important. In the case of an LLM for telematics causal analysis, a practitioner must choose a prompting method, select a model, and have confidence in the outputs. These decisions are arbitrary, and there are no comparative data to justify them. A poor strategy choice might produce plausible-looking but structurally inaccurate causal graphs, thus leading to inappropriate maintenance decisions or mistaken energy drivers.

Privacy is a crucial secondary consideration in this context. Telematics data are sensitive under the GDPR, and when entered into an LLM prompt, the model may reproduce or infer sensitive information in its output [24]. Whether prompt design can serve as a lightweight first-line control against this risk without requiring access to model internals is a question that the existing privacy literature has not addressed in this domain. This highlights a gap that previous studies have not addressed: how do causal discovery accuracies vary across different prompting strategies, and do the comparisons also show differences in privacy exposure?

1.4 Research Gap

Comparatively little attention has been paid to prompting and the quality of causal discovery outputs from LLMs. Studies use a single strategy and evaluate whether the output is usable; however, no study has systematically compared strategies against a quantitative ground truth in a telematics scenario. In addition to strategy comparison, the function of model design is ambiguous: do reasoning-focused models produce more accurate causal graphs than general-purpose models of the same or greater size?

One dimension that has not been investigated is the relationship between prompt design and sensitivity to leakage. Telematics data are privacy-sensitive under the GDPR, and there is a risk that when included in an LLM prompt, the model reproduces or infers sensitive information in its output. Whether this risk differs across prompting strategies and whether a well-designed prompt can lower it without sacrificing causal quality has not been studied in this domain [10].

Overall, these gaps leave two key questions unanswered: which prompting techniques produce the most accurate causal outputs on telematics data, and whether strategy choice also affects privacy as a secondary dimension. These problems are especially relevant in the vehicular domain, where data privacy obligations and operational decision quality depend on the answers.

1.5 Research Questions

This study is organized around three research questions that follow directly from the gaps identified above:

- **RQ1:** Which prompting strategies produce the most accurate causal discovery results when LLMs analyze vehicle telematics data?
- **RQ2:** How does the model architecture influence performance across different prompting strategies?
- **RQ3:** To what extent does prompt design affect sensitivity leakage as a secondary evaluation dimension, and can it serve as a lightweight privacy control?

These questions help establish prompt engineering as more than just a tool, but also as a design decision with implications for both analytical reliability and privacy in sensitive domains.

1.6 Contributions

This study makes four contributions to the intersection of prompt engineering, causal discovery, and vehicular data analytics.

The first is a structured comparative evaluation of five prompting strategies: Few-Shot, Chain-of-Thought, Style, ReAct, and Meta, applied to causal discovery on real EV telematics data across three open-weight large language models (LLMs): Llama 3 (8B), DeepSeek-R1 (8B), and Qwen (14B). The evaluation was designed to isolate the effect of prompt structure on causal graph quality, measured against a literature-grounded reference graph using eight structural metrics.

The second contribution is a causal quality evaluation framework that enables a systematic comparison of prompting strategies across models. By framing the task as causal graph discovery rather than open-ended analysis, this thesis provides a quantitative basis for strategy comparison that prior work in LLM-based telematics analytics has not established.

Third, this study introduces the Sensitivity Leakage Rate (SLR) as a secondary evaluation dimension. The SLR is a binary indicator that captures whether LLM outputs expose or enable the inference of sensitive information from the input prompt. This adds a privacy perspective to the causal quality comparison, asking not only which strategy performs best analytically but also whether it has implications for data exposure.

Finally, this thesis translates its findings into practical guidelines for deploying LLMs on sensitive telematics data by addressing which strategies and model architectures produce the most reliable causal insights while keeping sensitivity leakage under control.

2 Literature Review

2.1 Large Language Models and Causal Discovery

One of the most significant shifts in modern artificial intelligence is the rise of large language models. Earlier natural language processing systems were built around task-specific designs and hand-crafted features, which limited their generalizability. The Trans-

former introduced by Vaswani et al. [27] changed this by replacing sequential processing with self-attention mechanisms, allowing models to capture relationships across entire input sequences. Subsequent research has shown that scaling model size, training data, and computation leads to qualitative improvements in capability that do not appear in smaller models, a phenomenon often described as emergence [29].

Models such as GPT-4, LLaMA, and DeepSeek are trained on text corpora containing billions of tokens, which enables them to perform diverse linguistic and reasoning tasks. What sets them apart from earlier systems is not just their scale but the range of capabilities that can be elicited through prompting alone, without any task-specific retraining. These models can perform multi-step reasoning, process structured data, answer domain-specific questions, and produce coherent analytical outputs. This capability is known as in-context learning and emerges from pre-training; it is further refined by alignment techniques such as instruction tuning and reinforcement learning from human feedback (RLHF), which improve the consistency and reliability of model responses [15].

For the purposes of this thesis, what matters most is not the internal architecture of these models but their use as analytical tools. When structured or semi-structured data are processed through a prompt, large language models can surface patterns, flag anomalies, and generate summaries that would require considerable manual effort to produce. This makes them genuinely attractive for data-intensive domains, such as vehicle telematics. It also means that the prompt content and structure directly shape the model outputs, including sensitive information that it may expose or infer. How prompting affects both causal discovery quality and data sensitivity is both a performance question and a central concern of this study.

2.2 Prompt Engineering and In-Context Learning

Prompting has shifted from being a practical solution to a distinct field of research. Rather than retraining a model for each new task, users can direct it only with natural language instructions, a paradigm that has made LLMs far more accessible but has also introduced new sources of variability and risk.

Zero-shot prompting is considered the base prompting strategy, where only the task description is given to the model without any input-output examples, making the model rely solely on its learned knowledge to perform the task successfully. Few-shot prompting is an enhanced version of the zero-shot prompting strategy, in which input-output examples are added to the prompt to improve performance in structured tasks [4]. The Chain-of-Thought (CoT) prompting technique further improves the few-shot prompting technique by prompting the model to reason before arriving at the solution, which works well for multi-step tasks [29]. More recent strategies, such as ReAct, interleave reasoning with action steps. Meta-prompting and role-based prompting are models that provide explicit personas or high-level instructions that frame how the model should approach a task.

Strategies share the assumption that prompt structure affects model output in meaningful and reasonably predictable ways. Research has confirmed this, but it has also revealed a fragile relationship. The slightest change in wording, sequencing of examples, or even in the instruction itself could result in vastly different results generated by the same model [33]. Such sensitivity is both a strength and a weakness for the purposeful construction of prompts to improve model output, and also increases the risk that poorly designed prompts will produce unintended outputs, including the reproduction or

inference of sensitive information.

The latter has received comparatively little attention so far. Prompt engineering research mostly focuses on capabilities, enhancing accuracy, reducing hallucinations, and broadening the variety of tasks. However, the effect of prompt structure on the quality of structured analysis remains largely unexplored. Whether reasoning-focused techniques such as CoT and ReAct are better than simpler strategies such as few-shot on causal tasks is an open empirical question that this study directly addresses.

2.3 Privacy and Sensitivity Leakage in LLMs

Historically, privacy in machine learning has been considered a data protection problem. A primary concern is preventing models from leaking or reproducing sensitive information from their training data. This risk was substantiated when researchers demonstrated that language models can reproduce verbatim texts, including personal identifiers and private documents [5]. The research community has devised various countermeasures against this threat, focusing largely on mitigations performed during the training process. Differential privacy injects random noise into the gradient steps to bound the influence of each training sample on the model weights [9]. Data deduplication and filtering decrease the likelihood that sensitive information will enter the training corpus. Membership inference defense seeks to limit adversaries' ability to detect whether a particular record is included in the training dataset.

These countermeasures are technically mature and have genuine value. However, their limitation is structural: they are designed to protect information during or before training but not to control the output of a deployed model at inference time. Whatever the model generates at inference time is shaped by the full contents of the prompt in ways that are difficult to predict or to control.

Inference-time privacy issues are different from training-phase privacy problems. Duan et al. [8] have shown that the privacy risk of prompting, especially in the form of in-context learning, is significantly higher than that of fine-tuning since demonstration examples embedded in a prompt are directly exposed to the model at runtime and not absorbed through gradient updates. Empirical work has also demonstrated that LLMs can infer demographic features, behavioral tendencies, and even geographical information from contextually adjacent inputs that seem benign on their own [24].

The most comprehensive treatment of this problem is the survey by Edemacu and Wu [10], which classifies the privacy protection methods for LLM prompting into four categories: non-DP methods, including sanitization, obfuscation, and encryption [7]; local differential privacy at the word, sentence, or document level [11]; global differential privacy methods, such as PATE-based frameworks and DP synthetic data generation; and other scenarios, including federated learning and datastore protection.

None of these frameworks explore whether the prompt structure itself may prevent leakage in the absence of any formal privacy mechanism, which this study considers as a secondary assessment dimension in addition to the core causal quality comparison.

2.4 LLMs in Vehicle Telematics

Vehicle telematics systems produce some of the most information-dense data streams in consumer technology today. A modern connected vehicle continuously collects high-

frequency records across dozens of channels, including the engine state, energy consumption, speed and acceleration profiles, GPS coordinates, and diagnostic fault codes. These signals are technical and impersonal in isolation, but together they form a behavioral fingerprint specific enough to reveal where a driver lives and works, when they drive, and how they drive. This sensitivity has attracted legal attention: under the GDPR, location and behavioral data qualify as personal data, with specific obligations for how they must be handled, making privacy a non-negotiable consideration for any telematics implementation.

The volume and multidimensional structure of telematics data have driven interest in LLM-based analytical pipelines that can interpret patterns and provide natural language summaries without requiring users to write code. Early work has shown that LLMs can process structured time series inputs when organized suitably in a prompt [21], making them an attractive tool for operational telematics analytics.

LLMs and telematics data create an analytical pipeline with real implications for the quality of causal insights generated and the sensitivity of the information exposed, based on the choices made in the design. This junction has not been comprehensively studied in the telematics-LLM literature.

Table 1: Benchmark of related work on LLMs and AI for vehicle maintenance and telematics. This study is the only one to explicitly address privacy alongside LLM-based vehicle analytics.

Reference	**Objective**	**Target Variable**	**Methods**	**LLM**	**Prompt-level privacy?**
Pavlopoulos et al. (2024) [17]	Automotive fault nowcasting	Fault category / repair action	ML + NLP	No	No
Zhang et al. (2025) [30]	NEV fault information retrieval	Fault cause / solution	LLM + KG + RAG	Yes	No
Lin & Miao (2025) [12]	Vehicle fault diagnosis	Fault diagnosis / repair	KG + LLM	Yes	No
Mahale et al. (2025) [14]	AI-driven PdM review	Maintenance decision	Review of AI/ML PdM	Yes	No
Theissler et al. (2021) [25]	ML-based automotive PdM	Failure / maintenance need	ML-based PdM survey	No	No
Qaid et al. (2024) [18]	LLM fault diagnosis from sensor data	Machine fault class	FD-LLM	Yes	No
Chen et al. (2025) [6]	Large models for monitoring	Fault diagnosis / decision	LLMs + RAG + fine-tuning	Yes	No
Wang et al. (2023) [28]	PHM with local knowledge base	Maintenance insight	LLM + local KB + prompting	Yes	No
Lukens et al. (2024) [13]	PHM copilot for maintenance	Maintenance action	LLM agents	Yes	No
Palma et al. (2025) [16]	LLM-based predictive maintenance	Anomaly / failure risk	LLM + anomaly detection	Yes	No
Alsaif et al. (2024) [1]	Multimodal LLM fault detection	Fault detection / diagnosis	Multimodal LLM	Yes	No
Bao et al. (2025) [2]	Interpretable LLM fault diagnosis	Bearing fault class	CNN + fine-tuned LLM	Yes	No
Shi et al. (2025) [21]	LLMs for time-series analysis	Time-series prediction	LLM time-series methods	Yes	No
Shirole et al. (2025) [22]	Driver behavior scoring review	Driver behavior / performance	ML + DL + generative AI	Yes	No
Bitam et al. (2025) [3]	AIoT predictive maintenance review	Fault detection / RUL	AIoT + edge AI + digital twins	No	No
This study	**Comparative evaluation of prompting strategies for LLM-based causal discovery on EV telematics**	**Causal discovery DAG for predictive maintenance**	**Lightweight LLM + contextual + role-based prompting**	**Yes**	**Yes**

2.5 Research Gap and Positioning

The existing literature on LLMs for vehicle maintenance and telematics has grown substantially, including fault diagnosis, predictive maintenance, PHM copilots, and time-series analysis. What do these studies have in common? All of them, without exception, are about whether LLMs can provide valuable outputs, but not whether the prompting approach produces the most accurate or dependable outputs. Several already include built-in prompt engineering: Wang et al. [28] and Lukens et al. [13] both employ prompting to propose maintenance, but neither examines how strategy choice affects output quality. Causal discovery as an LLM evaluation task in telematics has not been conducted.

The second gap is privacy issues. None of the 15 studies in Table 1 considered whether prompt design influences the sensitive telematics data that the model reveals in its output. This study provides this dimension as a secondary evaluation of the fundamental causal quality comparison. This study aims to fill these gaps by comparing prompting strategies in causal discovery quality and how strategy choice impacts sensitivity leakage within the same framework.

3 Methodology

Figure 1 provides an overview of the experimental pipeline described herein. The methodology proceeds from data acquisition and preprocessing through ground truth construction, prompting experiments across three LLMs, and dual evaluation of causal discovery quality and sensitivity leakage.

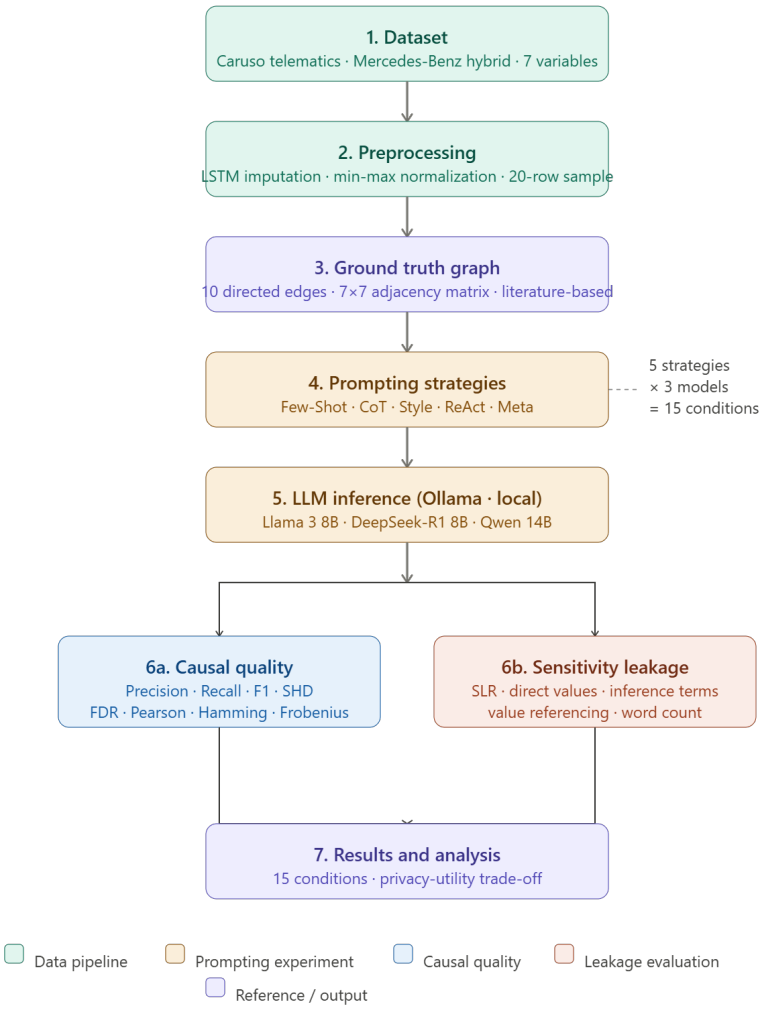


Figure 1: Overview of experimental methodology.

3.1 Dataset Description

The dataset for this study was provided through the Caruso data platform and consisted of car telematics records from a Mercedes-Benz hybrid vehicle with the vehicle unit RU043701, which was operated for a two-month period from March to April 2025. The data are time-stamped and include seven variables measured at regular intervals: battery voltage (`batteryvoltage`), state of charge high level (`sochighlevel`), average energy consumption (`energyconsumptionaverage`), remaining electric energy (`electricremaining`), total mileage (`mileage`), vehicle speed in km/h (`speed_kmh`), and acceleration in km/h^2 (`acceleration_kmh2`).

These variables define the energy state and dynamics of the vehicle. The battery voltage indicates that energy storage is available. The state of charge and remaining electric range are measures of efficiency and operational capacity. Driving behavior is characterized at the granular level by the mileage, speed, and acceleration. Individually, each feature may seem technical, but together they provide a behavioral profile specific enough to uncover regular usage habits, route preferences, and unique driving characteristics of

the driver.

Before the analysis, the dataset was normalized using min-max scaling to bring all variables into a common numerical range. A sample of 20 rows was selected from the cleaned and normalized dataset, combining rows from both the moving and stationary vehicle states to ensure sufficient variation in all variables. Table 2 presents an extract of the samples.

Table 2: Extract of the normalized telematics dataset sample (5 of 20 rows shown). All values were scaled to the range [0, 1]. The sample was selected to capture variation across both stationary and moving states.

Row	batteryvolt.	sochighlev.	energycons.	electricrem.	mileage	speed_kmh	accel_kmh2
1	0.1955	1.00	0.9814	0.8550	0.0000	0.0001	0.6694
2	0.2857	0.77	0.1512	0.7023	0.8525	0.0000	0.6694
3	0.3459	1.00	0.5703	0.8244	0.4958	0.0000	0.6694
4	0.3609	0.50	0.7825	0.4046	0.1923	0.0917	0.6797
5	0.3609	0.00	0.2520	0.0000	0.7448	0.0439	0.6431
...	...	...	...	...	...	...	...

3.2 Ground Truth Causal Graph

To enable the quantitative evaluation of the causal outputs generated by the LLMs, we constructed a reference causal graph based on well-established domain knowledge from the electric vehicle literature. The graph specifies ten directed causal relationships among the seven variables, backed by peer-reviewed sources on EV energy management, battery degradation, and driving dynamics [19, 32, 31, 20, 23].

The battery voltage directly affects the state of charge, and the voltage level is the major source of information on the remaining energy in a battery pack [19]. The state of charge influences the remaining electric energy and energy consumption; a higher charge level increases the available range, and aggressive driving drains it faster [32]. Higher consumption reduces the remainder of the electric energy and the available range [23]. Mileage is related to battery voltage through long-term degradation, which also affects energy consumption, and the accumulated wear reduces efficiency [19, 31]. Speed positively affects energy consumption owing to aerodynamic drag at high speeds and negatively affects residual electric energy at high speeds [20]; a higher speed depletes the battery faster. Changes in speed over a short period of time increase the power draw through acceleration, which immediately affects the energy consumption and speed [20].

This ground truth was represented as a 7×7 adjacency matrix with directed edges as 1 and non-existent edges as 0. Table 3 illustrates the adjacency matrix, with directed causal edges shown as bold 1 entries. The evaluation metrics specified in Section 3.5 were used to evaluate each LLM-generated causal graph.

Table 3: Ground-truth adjacency matrix. Entry $A_{ij} = 1$ (bold) indicates a directed causal edge from variable i (row) to variable j (column). BV = batteryvoltage, SOCH = sochighlevel, ECA = energyconsumptionaverage, ER = electricremaining, MIL = mileage, SPD = speed_kmh, ACC = acceleration_kmh2.

From \ To	BV	SOCH	ECA	ER	MIL	SPD	ACC
BV	0	**1**	0	0	0	0	0
SOCH	0	0	**1**	**1**	0	0	0
ECA	0	0	0	**1**	0	0	0
ER	0	0	0	0	0	0	0
MIL	**1**	0	**1**	0	0	0	0
SPD	0	0	**1**	**1**	0	0	0
ACC	0	0	**1**	0	0	**1**	0

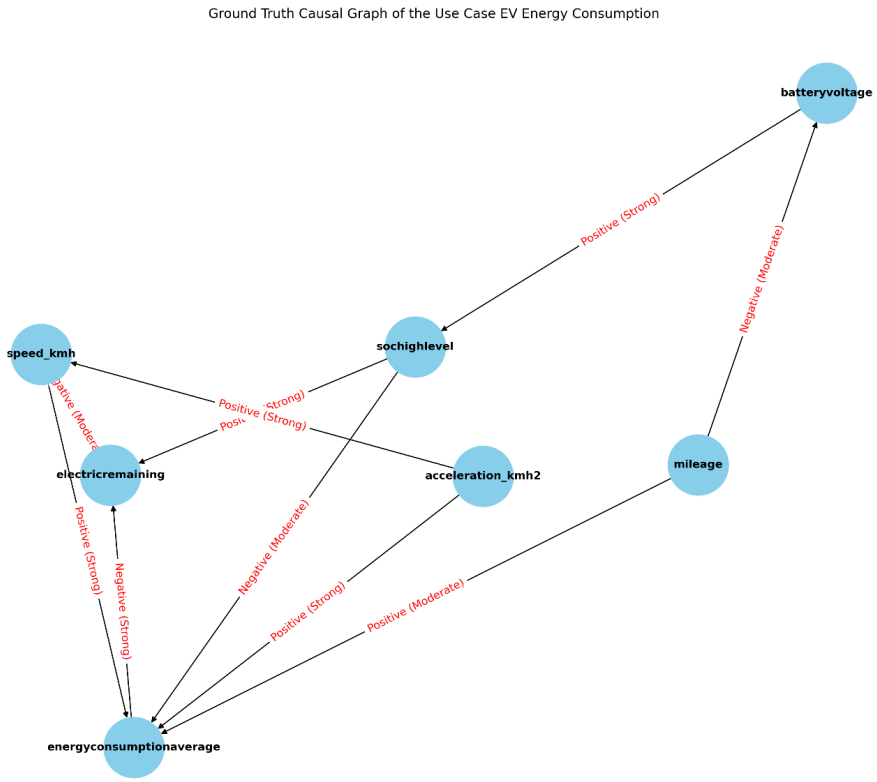


Figure 2: Ground truth causal graph showing the directed relationships among the seven EV telematics variables.

3.3 Prompting Strategies

The central experimental variable in this study was the prompting strategy used. The dataset, analytical work, and LLMs were the same across all conditions, with only the structure and framing of the prompt changing. Thus, the observed changes in output, including the variances in causal discovery quality and sensitivity leakage, can be attributed to prompt design rather than to differences in the data or model configuration.

We selected five prompting strategies, each representing a different way of structuring the interaction between the user and model. All five included the same base prompt, which included a 20-row dataset, variable names, and short domain descriptions. Only the framing layered on top of the image differs. Table 4 summarizes these five strategies.

Table 4: Summary of the five prompting strategies evaluated in this study.

Strategy	Description	Key characteristic
Few-Shot	Provides two causal notation examples before the task	Reduces output variability; may prime domain relationships
Chain-of-Thought	Step-by-step reasoning before final answer	Improves analytical quality; produces more verbose outputs
Style	Assigns expert automotive analyst persona	Role-based framing; effect varies significantly by model architecture
ReAct	Interleaves reasoning and action steps	Explicit process trace; verbose outputs create leakage surface area
Meta	Delegates strategy selection to the model itself	Can produce efficient or unpredictable outputs

3.3.1 Few-Shot Prompting

The Few-Shot prompt provides two concrete examples of the desired causal notation before the task. This decreases the variability of the output and offers the model a clear pattern to follow; the examples themselves may prepare the model to reference domain-specific relationships [4].

Example: Few-Shot Prompt

"Below are two examples of causal edges from prior datasets:
batteryvoltage $\rightarrow$ sochighlevel
speed_kmh $\rightarrow$ energyconsumptionaverage
Now, analyze the following telematics data and list all causal relationships in the same notation."

3.3.2 Chain-of-Thought Prompting

In Chain-of-Thought prompting, the model is asked to reason step-by-step before arriving at a conclusion [29]. This tends to increase the output quality of analytical tasks but also produces more verbose responses, which creates a surface area for potential data leakage.

Example: Chain-of-Thought Prompt

"Let's think step by step about which variables in this dataset causally influence others, and then list the causal relationships in the form: FeatureA $\rightarrow$ FeatureB."

3.3.3 Style Prompting

Style prompting assigns the character of an expert automotive analyst to the model. The idea is that role framing constrains the model's outputs to precise, domain-appropriate ones. In practice, its effect varies significantly depending on the model architecture, as observed in the experiments reported in Section 4.

Example: Style Prompt

"You are an expert automotive analyst specializing in electric vehicle telematics. Based on the data provided, identify the causal relationships among the variables in the form: FeatureA $\rightarrow$ FeatureB."

3.3.4 ReAct Prompting

ReAct prompting interleaves reasoning and action phases, making the model's analytical process explicit. Its verbosity makes it an intriguing case for examining leakage.

Example: ReAct Prompt

"For each candidate relationship, alternate between a thought (reasoning about whether the relationship is causal) and an action (decision to include or exclude). Conclude with the final list of causal edges in the form: FeatureA $\rightarrow$ FeatureB."

3.3.5 Meta Prompting

Meta prompting provides high-level instructions to the model on how to perform the task, and the model can choose a method. This can yield efficient but unpredictable outputs, depending on how the model understands an instruction.

Example: Meta Prompt

"Decide which analytical approach is most appropriate for identifying causal relationships in this telematics dataset, then apply it and list the resulting causal edges in the form: FeatureA $\rightarrow$ FeatureB."

3.4 Experimental Setup

The experiment was conducted locally using the Ollama framework, a tool that allows open-weight language models to run on consumer hardware without relying on external application programming interfaces (APIs). As the dataset contains sensitive telematics information, running models locally was a crucial practical necessity, as sending real vehicle data to a commercial API endpoint would be a privacy risk, contradicting the aims of this study.

Three language models were used in this study: Table 5 provides a summary of the models and their experimental conditions.

Table 5: Models used in the experiments. Each model was evaluated across all five prompting strategies, producing 15 conditions.

Model	Parameters	Architecture type	Developer	Conditions
Llama 3	8B	General instruction-tuned	Meta	5
DeepSeek-R1	8B	Reasoning-focused (internal CoT)	DeepSeek	5
Qwen	14B	General instruction-tuned	Alibaba	5
Total		**5 strategies $\times$ 3 models**		**15**

LLaMA 3 is a general-purpose instruction-tuned model from Meta and is a common baseline for open-weight LLM evaluation. DeepSeek-R1 is a reasoning-focused model that uses an internal chain-of-thought processing, which generally produces more analytically structured outputs. Qwen is a larger model from Alibaba, included to examine whether larger parameter count correlates with better causal discovery or different leakage behaviour.

All models were used in their standard instruction-following configurations without fine-tuning or modification. The temperature was set to 0.2 and top-p to 0.9 across all runs to maintain a consistent and near-deterministic generation. Each strategy-model condition was run ten times, and the reported metrics were averaged across the ten runs. The outputs were saved and aggregated for further evaluation.

3.5 Evaluation Framework

Each LLM output was evaluated along two dimensions: causal discovery quality and sensitivity leakage. Table 6 summarizes all evaluation metrics used in this study.

Table 6: Evaluation metrics used in this study. TP, true positives; FP, false positives; FN, false negatives. The SLR introduced in this study is a practical indicator of prompt-level leakage.

Metric	Formula / Definition	Interpretation
Precision	TP / (TP + FP)	Proportion of predicted edges that are correct. Higher is better.
Recall	TP / (TP + FN)	Proportion of ground truth edges recovered. Higher is better.
F1 Score	2 $\times$ Precision $\times$ Recall / (Precision + Recall)	Harmonic mean of Precision and Recall. Balanced indicator.
SHD	Edge additions + deletions + reversals	Structural distance from the ground truth. A lower score is better (0 = perfect).
FDR	FP / (TP + FP)	Proportion of predicted edges that are incorrect. Lower is better.
Pearson r	Correlation of flattened matrices	Linear association between the predicted and ground truth matrices.
Hamming	Proportion of differing elements	Element-wise structural distance between the matrices.
Frobenius	$\|\text{Predicted} - \text{Ground Truth}\|_F$	Overall matrix deviation. A lower value indicates a closer agreement.
SLR	1 if leakage detected, else 0	Proposed metric: binary indicator of prompt-level sensitivity leakage.

The causal discovery quality was assessed by comparing the adjacency matrix produced by each strategy with the ground-truth adjacency matrix. Precision, recall, and F1 were defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{1}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3}$$

Where TP denotes true positives, FP false positives, and FN false negatives. The Structural Hamming Distance (SHD) is the total number of edge additions, deletions, or reversals required to change the predicted graph into the ground truth, where lower values indicate closer agreement [26]. The False Discovery Rate (FDR) measures the ratio of incorrectly predicted edges. Other structural similarity measurements include the matrix-level Pearson correlation, Hamming distance and Frobenius norm.

To measure the sensitivity leakage across all 15 conditions, this study proposes a binary indicator called the Sensitivity Leakage Rate (SLR). There is no standard metric for prompt-level leakage in the literature or previous studies, as existing privacy evaluation frameworks focus on training-phase risks rather than inference-time exposure from user-supplied prompts. Hence, the SLR is offered as a realistic and transparent evaluation tool. We classify an output as SLR = 1 if any of the three detection checks return positive: direct value reproduction, where numerical values from the input data are reproduced in the output; sensitive inference, detected by keyword matching against a set of privacy-relevant terms, including “driver,” “location,” “route,” “pattern,” “behavior,” and “habit”; and value referencing, where numbers in the output fall within a tolerance of one unit from any value in the input sample.

Before the main experiment, a preliminary test was conducted using a modified prompt that expanded the original task to summarize the results and identify important values. This was meant to be a stress test of the SLR framework because if it could not detect leakage under these more liberal settings, the reliability of the framework would be zero. All five techniques evaluated on Llama 3 (8B) achieved an SLR of 1, demonstrating that the framework is capable of detecting leakage when it is present. The main experiment was subsequently repeated using the original limited prompt, allowing any variation in leakage to be attributed to the scope rather than the detection limit.

Token and word counts were also recorded as secondary measures of verbosity and analytical productivity, respectively.

4 Results

4.1 Overview

In this section, we present the results of the causal discovery experiment on three open-weight language models, DeepSeek-R1 8B, Llama 3.1 8B, and Qwen3 14B, under five

prompting strategies: Few-Shot, Chain-of-Thought (CoT), Style, ReAct, and Meta. Each strategy-model condition was run ten times, and all metrics were reported across the ten runs. The evaluation measures the similarity of the adjacency matrices generated by the model to a causal graph with ten edges using eight structural criteria. Sensitivity leakage is evaluated as a supplementary dimension using the Sensitivity Leakage Rate (SLR).

The key point is that Llama 3.1 is the only model that consistently parses causal graphs across all five strategies. DeepSeek-R1 was a stronger causal reasoner under structured scaffolding; it did not produce edges for any of the five thesis strategies in a single run. This finding is explained by DeepSeek-R1's thinking architecture, rather than a lack of causal knowledge. Qwen3 did not generate any edge under any approach. The difference in performance across prompting strategies was smaller than that between model architectures.

4.2 Ground Truth Reference

The ground truth causal graph defines ten directed edges among seven vehicle telematics variables derived from the EV literature. Battery voltage causes state of charge (BV→SOCH); state of charge causes both average energy consumption and remaining electric energy (SOCH→ECA, SOCH→ER); energy consumption causes remaining electric energy (ECA→ER); mileage causes battery voltage and energy consumption (MIL→BV, MIL→ECA); speed causes energy consumption and remaining range (SPD→ECA, SPD→ER); and acceleration causes both energy consumption and speed (ACC→ECA, ACC→SPD). A perfect result would reproduce all ten edges with Precision = Recall = F1 = 1.0 and SHD = 0. Table 7 lists all the edges.

Table 7: Ground truth causal graph: ten directed edges among seven EV telematics variables. A perfect output yields Precision = Recall = F1 = 1.0 and SHD = 0.

Source	**Target**	**Abbrev.**	**Causal Mechanism**
Battery voltage	State of charge	BV→SOCH	Voltage level is the primary indicator of remaining charge
State of charge	Energy consumption	SOCH→ECA	Higher charge state enables higher consumption rates
State of charge	Electric remaining	SOCH→ER	Charge level determines available driving range
Energy consumption	Electric remaining	ECA→ER	Greater consumption depletes remaining range
Mileage	Battery voltage	MIL→BV	Accumulated wear degrades voltage output over time
Mileage	Energy consumption	MIL→ECA	Drivetrain wear reduces energy efficiency
Speed	Energy consumption	SPD→ECA	Aerodynamic drag increases energy draw at higher speeds
Speed	Electric remaining	SPD→ER	Higher speed depletes available range faster
Acceleration	Energy consumption	ACC→ECA	Power demand spikes during acceleration events
Acceleration	Speed	ACC→SPD	Acceleration directly produces change in speed

4.3 Causal Discovery Results

4.3.1 DeepSeek-R1 8B

DeepSeek-R1 produced zero parseable causal edges for all five thesis strategies: Few-Shot, CoT, Style, ReAct, and Meta. Therefore, F1 = 0.000 and SHD = 10 for all the conditions. The model did respond, however, as indicated by inference latencies from 29 to 44 seconds, yet the raw text field saved by the evaluation pipeline was empty in all cases.

This appears to be due to the reasoning-oriented architecture of DeepSeek-R1, a model trained with reinforcement learning on chain-of-thought traces, so much of its reasoning is done internally in terms of think tokens, which are stripped by the Ollama pipeline before saving the final output. Under the five thesis strategies, the model appears to complete its reasoning internally without emitting a structured edge list, leaving the parser with empty strings and zero extracted edges.

Importantly, this does not imply the absence of causal reasoning capabilities. DeepSeek-R1 with Step-Back Prompting, which explicitly scaffolds abstraction before edge generation, obtained F1 = 0.750 with Precision = 1.000 and SHD = 4, recovering six out of ten ground-truth edges without any false positives. Additional evaluations conducted outside the five thesis strategies further support this: F1 = 0.500 under Contextual Prompting and F1 = 0.400 under Instruction Prompting. These results suggest that DeepSeek-R1 produces high-quality causal graphs when provided with prompts that provide a concrete framework that matches its output structure.

4.3.2 Llama 3.1 8B

For all five strategies, Llama 3.1 produced non-zero F1 scores and was the only model in this experiment that consistently produced parseable causal graphs. Its response latencies ranged from 3.1 to 5.9 seconds, which is much shorter than those of DeepSeek-R1 and consistent with its general instruction-tuned design that outputs a direct answer instead of a long reasoning trace.

Chain-of-Thought achieved the highest F1 of 0.375 with Precision = 0.500 and Recall = 0.300. It produced six predicted edges, three of which were true positives, and three were false positives (SHD = 10). The reasoning structure seems to encourage the model to propose more relations, improving recall but also inserting incorrect edges on top of the correct ones.

Meta Prompting was the top performer in terms of precision with Precision = 1.000, Recall = 0.200 and F1 = 0.333. The model only produced two edges, BV$\rightarrow$SOCH and SOCH$\rightarrow$ER, and both were true positives from the ground truth, achieving perfect precision at SHD = 8. Llama 3.1 adopted a conservative approach when selecting its analytical technique, committing only to the two relationships it was most certain of. The Meta condition is characterized by a precision-recall trade-off, which in turn has a higher confidence per edge but reduced coverage.

The F1, Precision, Recall, and SHD for Few-Shot, Style, and ReAct were nearly the same: 0.250, 0.333, 0.200, and 12, respectively. Each strategy resulted in the model proposing six edges, two of which were true positives. The convergence of three structurally different strategies–one providing examples, one assigning a persona, and one interleaving reasoning and action steps on the same numerical output–indicates that Llama

3.1 defaults to a similar internal causal reasoning process regardless of how the prompt is framed when no explicit step-by-step direction is provided. Providing demonstration examples (Few-Shot) fails to support the model better for this task compared to role assignment (Style) or action-interleaving (ReAct).

4.3.3 Qwen3 14B

Qwen3 failed to produce any parseable edges for all the five strategies with F1 = 0.000 and SHD = 10 for all the conditions. Qwen3's empty outputs have far longer latencies–71 to 105 seconds–suggesting the model created vast amounts of text, unlike DeepSeek-R1. Checking the raw output confirms this: Qwen3 produced verbose natural language explanations of relations between variables. The edge parser did not identify any structured "variable -¿ variable" notation in any of the explanations. The model engaged with the task conceptually but did not produce the graph notation.

Qwen3 has 14 billion parameters, making it the largest tested model. Most directly, the failure to consistently provide structured causal output across all strategies demonstrates that the parameter-count-based assumption of causal reasoning capabilities is questionable, while both 8B models produced at least some parseable results under proper conditions. The difference between Qwen3 and Llama 3.1 is only due to the alignment of the output format, not the model size.

Table 8 presents the full results for all 15 conditions.

Table 8: Causal discovery results for all 15 strategy-model conditions (averaged across 10 runs per condition). Bold F1 values indicate the best result within each model. SHD: Structural Hamming Distance; FDR: False Discovery Rate. Pearson r is shown as '–' when both predicted and ground truth matrices produce zero variance (undefined correlation).

Model	Strategy	Prec.	Rec.	F1	SHD	FDR	Pearson r	Frob.
DeepSeek-R1 8B	Few-Shot	0.000	0.000	0.000	10	0.000	–	3.162
DeepSeek-R1 8B	CoT	0.000	0.000	0.000	10	0.000	–	3.162
DeepSeek-R1 8B	Style	0.000	0.000	0.000	10	0.000	–	3.162
DeepSeek-R1 8B	ReAct	0.000	0.000	0.000	10	0.000	–	3.162
DeepSeek-R1 8B	Meta	0.000	0.000	0.000	10	0.000	–	3.162
Llama 3.1 8B	Few-Shot	0.333	0.200	0.250	12	0.667	0.120	3.464
Llama 3.1 8B	CoT	0.500	0.300	**0.375**	10	0.500	0.274	3.162
Llama 3.1 8B	Style	0.333	0.200	0.250	12	0.667	0.120	3.464
Llama 3.1 8B	ReAct	0.333	0.200	0.250	12	0.667	0.120	3.464
Llama 3.1 8B	Meta	1.000	0.200	**0.333**	8	0.000	0.407	2.828
Qwen3 14B	Few-Shot	0.000	0.000	0.000	10	0.000	–	3.162
Qwen3 14B	CoT	0.000	0.000	0.000	10	0.000	–	3.162
Qwen3 14B	Style	0.000	0.000	0.000	10	0.000	–	3.162
Qwen3 14B	ReAct	0.000	0.000	0.000	10	0.000	–	3.162
Qwen3 14B	Meta	0.000	0.000	0.000	10	0.000	–	3.162

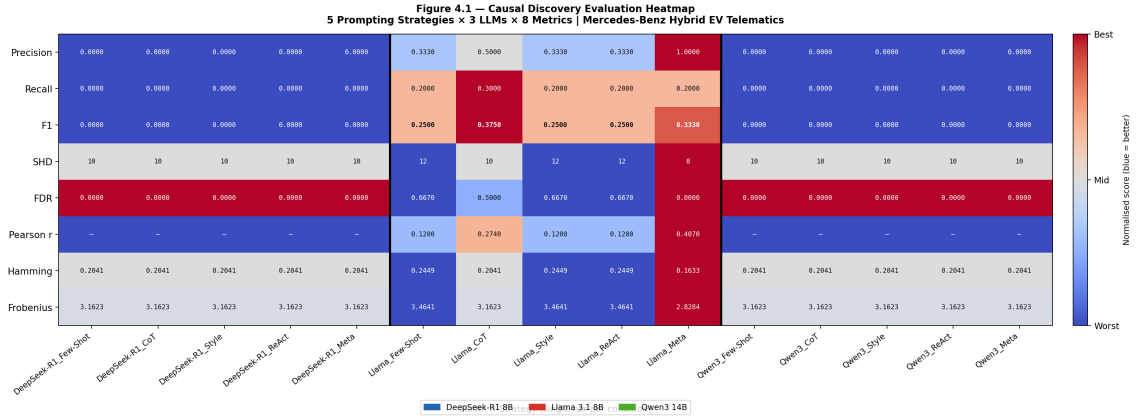


Figure 3: Causal Discovery Evaluation Heatmap. 5 Prompting Strategies $\times$ 3 LLMs $\times$ 8 Metrics on the Mercedes-Benz Hybrid EV Telematics dataset.

4.4 Cross-Model Comparison

Llama 3.1 is the strongest model across the five thesis strategies. It had a non-zero F1 in all five conditions, with an average F1 of 0.292 and a range of 0.250–0.375. The average F1 of DeepSeek-R1 and Qwen3 was 0.000 using the same five methods. This makes Llama 3.1 the only viable model for the five-strategy causal discovery.

The difference between DeepSeek-R1 and Qwen3 is informative. DeepSeek-R1 exhibited promising causal reasoning abilities with the correct prompting framework. Its Step-Back result of F1 = 0.750 and Precision = 1.000 outside the thesis scope confirms this finding. The five thesis strategies do not activate this latent capability. In contrast, Qwen3 generates verbose natural-language outputs without producing any structured graphs regardless of the strategy, showing that the mismatch is more fundamental: it is a formatting and alignment issue that no thesis strategy resolves.

The strongest evidence against the parameter count hypothesis is that Llama 3.1 at 8B outperforms Qwen3 at 14B across every metric. The design of the model and how it is trained and fine-tuned matter more for performance on this task than the raw scale.

No single method was dominant over all the models. CoT achieved the highest F1 for Llama 3.1, but 0 for DeepSeek-R1 and Qwen3. Meta achieved perfect precision for Llama 3.1 but zero output from the other two models. This indicates that the impacts of strategies are model-dependent. For models that fail at the parsing level, the choice of strategy does not matter.

4.5 Reliability of Multi-Run Results

In this experiment, each condition corresponded to ten LLM calls at a temperature of 0.2, and the reported metrics were averaged across those ten runs. Each condition was run ten times to determine the influence of output variability on the reported figures and to provide a more stable basis for comparison across strategies and models.

Although a temperature of 0.2 reduces the stochasticity, it does not completely remove the variation in the generated edge sets. For Llama 3.1 conditions with non-zero F1 scores, slight changes between runs would be expected; a condition with six edges in one run might generate five or seven in another. The same F1 = 0.250 achieved with

Few-Shot, Style, and ReAct prompts implies some stability but should not be construed as a strong estimate of predicted performance across repeated runs.

The null results for DeepSeek-R1 and Qwen3 seem to be systematic rather than stochastic. DeepSeek-R1 consistently returned empty parseable outputs across all five thesis strategies, even though inference latencies ranged from 29 to 44 seconds per condition (suggesting that the model performed properly but did not generate structured edge lists). This pattern is consistent across numerous prompting strategies, indicating an architectural issue rather than a random sample variation.

4.6 Sensitivity Leakage Results

In this study, we propose the Sensitivity Leakage Rate (SLR) as a binary measure of sensitivity leakage, defined by three detection criteria: direct numerical reproduction of input values, appearance of privacy-relevant inference terms such as “driver,” “location,” “route,” “pattern,” and “habit,” and output values within one unit of any input number. An SLR of 0 indicates that none of the criteria were detected.

With a constrained causal-edge output format, all strategy-model conditions produced SLR = 0. It appears that requesting answers in structured “variable -¿ variable” notation reduces both numerical reproduction and discursive language that can carry sensitive information. Even Qwen3, which produced explanatory text instead of edge lists, did not trigger any of the leakage criteria, suggesting that abstract causal discussion alone does not constitute the forms of exposure targeted by the SLR framework.

A validation experiment confirmed that the framework works: we modified the prompt to request data summaries and notable values alongside causal edges, which produced an SLR of 1 across all five strategies under Llama 3.1. This is important because it rules out the possibility that the null leakage result was simply a failure of the metric. The difference between these two sets of results points to one clear conclusion: what drives leakage is not strategy alone but how much latitude the prompt gives the model, specifically, whether it invites open-ended analytical prose or keeps the output tightly scoped.

4.7 Privacy-Utility Analysis

Figure 4 plots the F1 score on the horizontal axis and SLR on the vertical axis for all conditions. All points are on the SLR = 0 baseline, scattered across F1 values ranging from 0.000 to 0.375. In this experiment, the trade-off between privacy and utility expected by the literature, where stronger analytics come at the expense of more exposure of information, did not materialize. All Llama 3.1 strategies have situations in which there is excellent causal quality and zero leakage. There are also conditions for which there is zero causal output and leakage (DeepSeek-R1 and Qwen3). In this experimental context, we do not observe any tension between utility and privacy under the restricted output format used here.

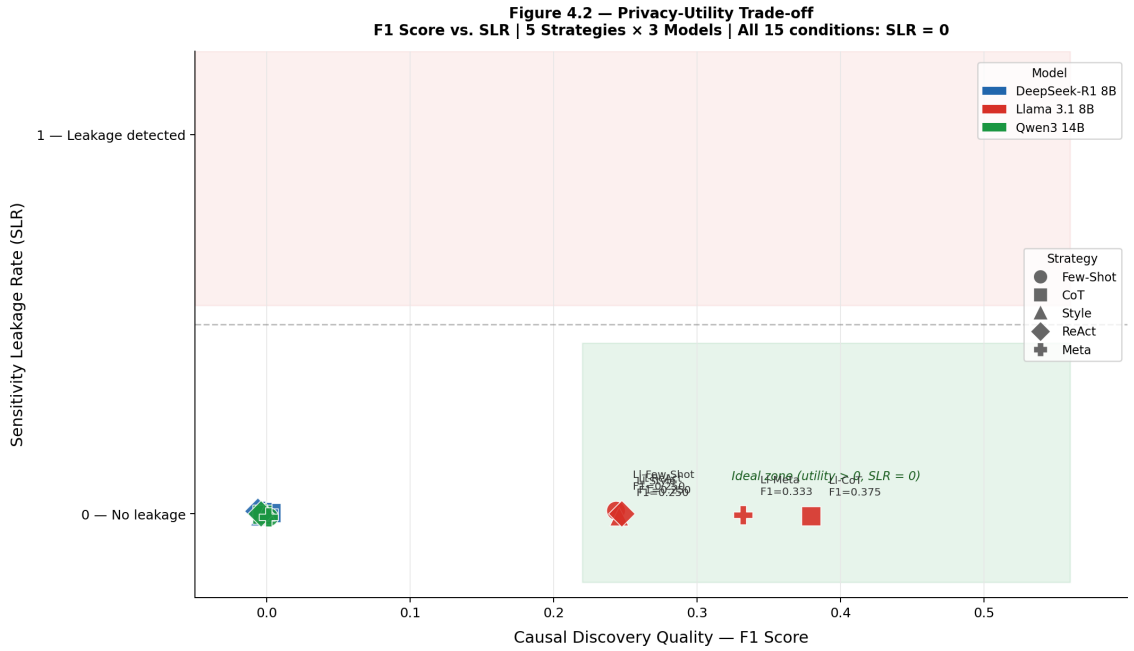


Figure 4: Privacy-Utility Trade-off. F1 Score vs. SLR across 5 strategies × 3 models. All 15 conditions achieve SLR = 0.

5 Discussion

5.1 Summary and Interpretation of Findings

This study compared five prompting strategies across three open-weight LLMs for the task of causal graph discovery from EV telematics data, with sensitivity leakage evaluated as a secondary dimension. The results clearly address the three research questions, given the single-dataset approach and the breadth of the conclusions.

RQ1 investigated which prompting strategies produced the most accurate causal outcomes. For Llama 3.1, CoT achieved the highest F1 score of 0.375, and Meta Prompting achieved the highest precision of 1.000. However, the differences across the five strategies were modest. DeepSeek-R1 and Qwen3 generated no parseable output under any thesis condition, while Llama 3.1's performance ranged between F1 = 0.250 and 0.375. The main finding for RQ1 was that model architecture plays a more significant role than strategy choice, and prompt design alone cannot compensate for incompatible output formatting.

RQ2 examined how the model architecture affected the causal reasoning quality. Architecture was the primary factor in the study. Llama 3.1 8B consistently outperformed Qwen3 14B across all thesis strategies despite having fewer parameters, demonstrating that scale alone does not predict performance in structured causal graph generation tasks. DeepSeek-R1 8B achieved the strongest causal results under Step-Back Prompting with F1 = 0.750 and Precision = 1.000, which only emerged with prompts that explicitly scaffolded structured output generation. For this task, training objectives and output format alignment are more important than the parameter count.

RQ3 asked whether prompt design serves as a lightweight privacy control. The Sensitivity

Leakage Rate (SLR) was 0 across all strategy-model conditions. The constrained edge-notation format appears to prevent both numerical reproduction and privacy-relevant analytical prose without an explicit privacy-preserving mechanism. We also conducted a validation experiment confirming that the framework detects leakage when prompts invite open-ended data summaries, which suggests that the null leakage result is genuine and not a failure of the metric. In this study, analytical utility and zero measured leakage coexisted in the constrained causal-discovery format.

5.2 Model Architecture and Causal Discovery Quality

The main empirical contribution of this study is the difference between the three models. Llama 3.1 consistently generated structured causal edge lists across all prompting strategies, reflecting strong instruction alignment with concise task-appropriate outputs. Its relatively low latency of 3–6 seconds per condition further suggests that the model directs responses rather than extended reasoning traces, which aligns well with the single-turn edge-list evaluation framework.

DeepSeek-R1 exhibited a fundamentally different behaviour profile. The model is trained using reinforcement learning on chain-of-thought reasoning traces, making it highly capable when prompts explicitly scaffold deliberative reasoning, followed by structured output generation. Step-Back Prompting activates this capability by first asking the model to abstract domain principles before committing to the causal edges. Under Step-Back, DeepSeek-R1 achieved the strongest overall performance in the study.

The five thesis strategies did not provide this form of scaffolding. Few-shot, style, ReAct, CoT, and meta-prompting either supplied examples, assigned roles, delegated reasoning, or structured intermediate actions, but none explicitly guided the transition from internal reasoning to final graph notation. As a result, DeepSeek-R1 produced long inference latencies of 29–44 seconds while returning empty parseable outputs. The evidence suggests that the model completed substantial internal reasoning without producing the structured edge list required by the evaluation pipeline.

This distinction is practically important. DeepSeek-R1 is not a weak causal reasoner, and its performance depends heavily on prompts that explicitly structure the final output. Prompting strategies that separate abstraction from structured answer generation appear to be particularly effective for reasoning-oriented architectures.

Qwen3 failed for a different reason. Although the model generated lengthy responses with latencies between 71 and 105 seconds, its outputs primarily consisted of explanatory natural language rather than parseable graph notation. The issue was not causal reasoning quality itself, but insufficient alignment with the strict “variable -¿ variable” output format required by the parser. Unlike DeepSeek-R1, whose capability can be unlocked through additional scaffolding, Qwen3 has a broader incompatibility with the structured-output requirements of the task.

5.3 Strategy Comparison Insights

Among the five thesis strategies, CoT and Meta Prompting produced meaningfully different behaviors for Llama 3.1. Few-Shot, Style, and ReAct converged to identical performance metrics. This convergence suggests that these prompt structures do not substantially alter default reasoning for causal graph discovery.

CoT achieved the strongest balanced performance for Llama 3.1, with F1 = 0.375. Explicit step-by-step reasoning instructions encouraged the model to explore additional causal pathways and increased recall from 0.200 to 0.300 while maintaining comparable precision, and SHD improved from 12 to 10.

Meta Prompting produced the highest precision in the dataset, achieving Precision = 1.000 with Recall = 0.200. Under this strategy, Llama 3.1 generated only two edges, both of which were true positives. This suggests that when the model is left to determine its own approach, it plays it safe, committing only to the relationships it feels most confident about rather than reaching for broader coverage.

The practical implication is straightforward: strategy choice should depend on what matters most for the application. If broader causal coverage is the priority and some false positives are acceptable, CoT is the better option. If high-confidence edges matter more than completeness, Meta Prompting is a stronger choice.

5.4 Relation to Existing Literature

The findings of this study reinforce a broader methodological issue in LLM evaluation: the measured performance depends not only on model capability but also on whether the evaluation protocol successfully activates that capability. DeepSeek-R1's strong performance under Step-Back Prompting and its null outputs under the five thesis strategies illustrate how prompt structure can determine whether the model's reasoning ability is actually surfaced in the output.

The convergence of the Few-Shot, Style, and ReAct results for Llama 3.1 is consistent with the observations of Errica et al. that some tasks exhibit high prompt sensitivity, while others show near-zero variation across prompting styles. In this study, these three strategies fell into the latter category, indicating limited marginal value from the additional prompt-engineering complexity for this model-task combination.

The SLR findings also align with the theoretical argument of Edemacu and Wu [10] that leakage risk is driven more by prompt scope than by the prompting strategy itself. Constraining outputs to edge notation suppressed leakage across all conditions, and a modified prompt requesting open-ended summaries consistently triggered leakage detection. Although the SLR metric remains preliminary and student-proposed, the directional finding is consistent across the experiments.

More broadly, none of the studies reviewed in Table 1 evaluated causal graph discovery from telematics data using quantitative graph metrics against a predefined ground truth. This study contributes a novel benchmark setting and provides evidence that architecture and output alignment are more important determinants of structured performance than parameter scale alone.

5.5 Practical Implications

For practitioners applying LLMs to causal discovery tasks using vehicle telematics data, model selection is the most important practical consideration. Llama 3.1 8B was the most reliable model for producing consistent and parseable causal graphs across multiple prompts. CoT is the preferred default strategy when balanced recall is required, whereas Meta Prompting is better suited for applications with high-confidence edge identification and minimal false positives.

For DeepSeek-R1 users, the prompts should explicitly scaffold structured reasoning and final output generation. Step-Back Prompting achieved F1 = 0.750 with Precision = 1.000, outperforming all Llama 3.1 results in this study. The key practical lesson is that reasoning-oriented models benefit from prompts that separate conceptual reasoning from final, structured responses.

From a privacy perspective, constrained edge-only output formatting appears to function as an effective, first-line leakage control. This approach requires neither model modification nor additional privacy infrastructure, yet produces SLR = 0 across all evaluated conditions. Whether this behavior remains robust under adversarial prompting or with more sensitive datasets remains an open question.

5.6 Limitations

First, the experiment was performed on a single dataset from a Mercedes-Benz hybrid vehicle over two months, provided by the Caruso platform. The results may not be directly applicable to other vehicle types, fleet-scale datasets, telematics systems, or longer observation periods.

Second, each condition was run ten times at a temperature of 0.2, and the reported metrics were averaged across those runs. While this reduces stochastic variability compared to a single run, a temperature of 0.2 does not eliminate output variation entirely, and the averaged metrics for Llama 3.1 conditions near the threshold of producing or omitting specific edges should be interpreted with this in mind.

Third, the ground-truth causal graph was drawn from the domain literature rather than experimentally validated by intervening in the dataset itself. The benchmark reflects informed expert assumptions about the EV system rather than a directly verified causal structure.

Finally, the SLR metric is a student-proposed framework that has not been validated against established privacy or auditing methodologies. It captures only three predefined forms of leakage and does not measure exposure severity, re-identification risk, or adversarial extraction capability. Privacy-related conclusions should be treated as exploratory findings, rather than definitive guarantees.

6 Conclusion

This study addressed two questions that the existing literature has not tackled together: how prompting strategy affects the quality of causal discovery outputs on telematics data, and whether strategy choice also interacts with sensitivity leakage. Both questions are important because LLMs are increasingly being used to process sensitive operational data that carries real obligations under frameworks such as the GDPR.

The methodological contribution is a dual evaluation framework that operates across five prompting strategies and three open-weight LLMs. Causal quality is measured against a ten-edge literature-grounded reference graph using eight structural metrics, and sensitivity leakage is captured through the Sensitivity Leakage Rate, a binary indicator introduced in this study.

The main empirical findings are clear and unambiguous. Llama 3.1 8B was the only model that consistently parsed causal graphs across all five strategies, achieving the

highest F1 of 0.375 under CoT and achieving perfect precision on a conservative two-edge prediction under Meta Prompting. DeepSeek-R1 8B produces no structured output under any of the five thesis strategies because its reasoning is internal and never gets emitted into an edge list; however, under Step-Back Prompting outside the thesis scope it achieves F1 = 0.750 with Precision = 1.000, the strongest result in the dataset. Qwen3 14B produced no parseable output under any strategy, confirming that parameter count alone does not guarantee structured causal graph generation. SLR was 0 across all conditions, and the anticipated privacy-utility trade-off did not materialize under the constrained edge-notation format.

In sum, this thesis demonstrates that for the specific task of LLM-based causal discovery on EV telematics data, three observations stand out. First, model architecture and output-format alignment matter more than parameter count or prompt sophistication alone. Second, reasoning-oriented models such as DeepSeek-R1 can deliver strong causal performance when prompts explicitly scaffold the transition from internal reasoning to structured output. Third, constrained output formats appear to function as an effective first-line privacy control under the SLR framework introduced here.

Future work should validate these findings across additional datasets and vehicle types, investigate whether the SLR framework generalizes under adversarial prompting, and explore prompting strategies that explicitly scaffold structured output generation for reasoning-oriented models. Extending the comparison to commercial, closed-weight models and to fleet-scale datasets would also help clarify the boundary between architecture-driven and scale-driven performance differences identified in this study.

Declaration on the Use of Artificial Intelligence Tools

I hereby declare that artificial intelligence (AI) tools were used responsibly in the preparation of this thesis. All AI-generated content has been critically reviewed, verified, and edited by me to ensure its accuracy, academic integrity, and alignment with the research objectives.

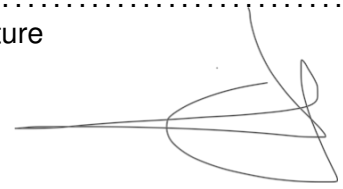
AI Tool	**Purpose of Use**
ChatGPT	Used to improve the clarity, grammar, and readability of the thesis text.
Grammarly Pro	Used for grammar checking, spelling correction, and stylistic improvement.
DeepL Translator	Used to support translation and language refinement between English and German where necessary.
Claude Sonnet	Used to produce simulation code and assist with data analysis scripts.

No AI system was used to generate or falsify data, conduct analysis, or produce content presented as original research.

I take full responsibility for the accuracy and integrity of all content in this thesis.

19.05.2026

Date, Signature



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A Telematics Variables and Abbreviations

The seven core variables used in this study are summarized below.

Table 9: Telematics variables used in the experiment.

Variable	Abbreviation	Description
batteryvoltage	BV	Instantaneous voltage of the battery pack
sochighlevel	SOCH	State of charge (high level), expressed as a percentage
energyconsumptionaverage	ECA	Average energy consumption
electricremaining	ER	Remaining electric energy in the battery
mileage	MIL	Total distance travelled by the vehicle
speed_kmh	SPD	Current speed in km/h
acceleration_kmh2	ACC	Current acceleration in km/h^2

B Full Adjacency Matrix Notation

The ten directed edges in the ground truth causal graph correspond to the following entries in the 7×7 adjacency matrix:

- BV → SOCH
- SOCH → ECA
- SOCH → ER
- ECA → ER
- MIL → BV
- MIL → ECA
- SPD → ECA
- SPD → ER
- ACC → ECA
- ACC → SPD

A perfect output reproduces all ten edges with Precision = Recall = F1 = 1.0 and SHD = 0.